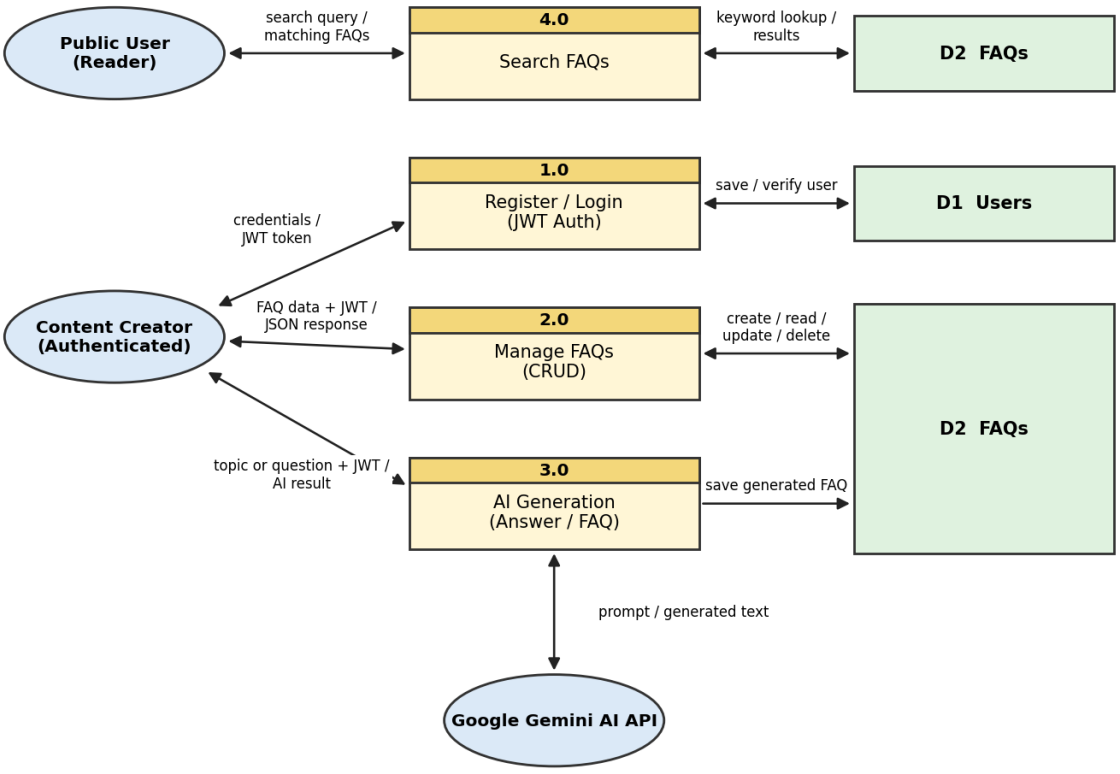
Data Flow Diagram

Date	16 September 2026
Team ID	SWTID-2026-3976
Project Name	AI FAQ Assistant API (AnswerBot)
Maximum Marks	2 Marks

Data Flow Diagram

The DFD shows how data moves between external entities, processes and data stores in the AI FAQ Assistant API. External entities are the Reader, the authenticated Content Creator and the Google Gemini API. Data stores are D1 Users and D2 FAQs (MongoDB collections).

DFD Level-1: AI FAQ Assistant API



Process	Description
1.0 Register / Login	Validates credentials, hashes passwords with bcrypt, stores users in D1 and returns a signed JWT.
2.0 Manage FAQs	JWT-protected create, update and delete of FAQ documents in D2; public read access.
3.0 AI Generation	Sends the topic or question to Google Gemini and returns a generated answer or FAQ pair, which can be saved to D2.
4.0 Search FAQs	Public keyword search over question, answer and category returning matching FAQs as JSON.